

DR. JITENDRA KUMAR



RESEARCH INTERESTS

Applied Machine Learning
Cloud Computing
Evolutionary Computing
Optimization

EDITORIAL SERVICES

- Scientific Reports, Springer
- Computer Communications, Elsevier
- Computer Networks, Elsevier
- Computers and Electrical Engineering, Elsevier
- Journal of Biomedical and Health Informatics, IEEE

RESEARCH INDEX (SINCE 2021)

- 🌐 Google Scholar 
 - Citations: 2608
 - h-index: 24
 - i-index: 31
- 🌐 Scopus 
 - Citations: 1607
 - h-index: 19
- 🌐 Orcid 

PERSONAL INFORMATION

Citizenship: Indian
Languages: English, Hindi

CONTACT DETAILS

@ [jitendrak\[at\]manit\[dot\]ac\[dot\]in](mailto:jitendrak[at]manit[dot]ac[dot]in)
☎ +91-9650932489
✉ MBC, MANIT Bhopal, India

BRIEF BIO

I, Dr. Jitendra Kumar serve as an Assistant Professor at an institution of national importance, India (Maulana Azad National Institute of Technology Bhopal, India), within the Department of Mathematics, Bioinformatics, and Computer Applications. My academic journey includes a Ph.D. from the National Institute of Technology, Kurukshetra, and prior experience as an Assistant Professor at the National Institute of Technology, Tiruchirappalli. My research interests include machine learning, computational intelligence, predictive analytics, and cloud computing.

A significant portion of my work addresses the challenges of workload forecasting in cloud data centers. Accurate workload prediction is vital for efficient resource management, ensuring that computational resources are allocated optimally to meet demand without over-provisioning. One of my work explored the integration of artificial neural networks with adaptive differential evolution techniques. This hybrid approach aimed to improve the accuracy of workload predictions by combining the learning capabilities of neural networks with the optimization strength of evolutionary algorithms. The resulting model exhibited enhanced performance in forecasting dynamic workloads, contributing to more responsive and efficient cloud resource management. I also proposed an ensemble learning-based predictive framework that integrates multiple machine learning models to forecast VM resource requests. This ensemble approach leverages the strengths of individual models, resulting in more robust and accurate predictions, which are critical for maintaining the performance and reliability of cloud services. My research also delves into the optimization of virtual machine (VM) placement within cloud data centers. Efficient VM placement is crucial for maximizing resource utilization and minimizing operational costs. He developed a secure and multi-objective VM placement framework that considers factors such as security requirements, resource constraints, and energy efficiency. This comprehensive approach ensures that VMs are allocated in a manner that balances performance with security considerations, thereby enhancing the overall reliability of cloud services.

Beyond research endeavors, I am actively involved in the academic community. I have published a significant number of research articles published in reputable journals and conferences, contributing to the advancement of knowledge in my areas of expertise. My collaborations with other scholars have led to interdisciplinary research that addresses complex challenges in cloud computing and machine learning. My work has received significant citations, reflecting the impact and relevance of my contributions to the fields of machine learning and cloud computing. This recognition underscores the importance of my research in addressing real-world challenges and advancing technological innovation.

To summarize, my research portfolio showcases a commitment to advancing the fields of machine learning, cloud computing, and predictive analytics. The proposed approaches to workload forecasting, resource optimization, and adaptive learning have contributed to more efficient and secure cloud data center operations.

POSITIONS HELD

ASSISTANT PROFESSOR

May 2024 - Present

Maulana Azad National Institute of Technology Bhopal, India

- ◇ Teaching: Neural Networks, Computer Architecture, and Computer Architecture & System Software.
- ◇ Administration: Time Table Coordinator, Department Academic Committee Co-Coordinator, Audit Verification Officer, Department Social Media Coordinator, Anti Ragging Committee Member.

ASSISTANT PROFESSOR

May 2020 - April 2024

National Institute of Technology Tiruchirappalli, India

- ◇ Teaching: Data Analytics, Big Data Analytics, Introduction to Programming, and Object Oriented Analysis and Design.
- ◇ Administration: Performance Analysis Committee Chairman, NBA Accreditation Committee Co-coordinator (MCA), Syllabus Revision Committee Co-coordinator (MCA), Coordinator Web Development, Institutional Development & Alumni Relations Data Coordinator (CA Department), Verification Officer NIMCET-2021, Accommodation & Space Allocation Committee.

ASSISTANT PROFESSOR

July 2019 - May 2020

GLA University Mathura, India

- ◇ Teaching: Introduction to Machine Learning, Design and Analysis of Algorithms, Cloud Computing, and Computer Organization.
- ◇ Administration: Student Performance Monitoring Committee Member.

ASSISTANT PROFESSOR

April 2014 - Oct 2015

Galgotias University, Greater Noida, India

- ◇ Teaching: Computer Programming & Problem Solving, Computer Graphics, and Universal Human Values & Ethics.
- ◇ Administration: Time Table, Result & Analysis, Anti Ragging Coordinator.

SOFTWARE ENGINEER

Nov 2013 - March 2014

Dayalbagh Educational Institute, Agra, India

- ◇ Assisted in developing the Online Admission System (<https://www.dei.ac.in/dei/admission/>).

TEACHING ASSISTANT

July 2011 - May 2013

Dayalbagh Educational Institute, Agra, India

- ◇ Assisted Dr. Lotika Singh to develop course notes, assignments, and lab manuals for Parallel Programming, Operating Systems, and Introduction to Linux.

EDUCATION

DOCTOR OF PHILOSOPHY

Oct 2015 - Feb 2020

National Institute of Technology Kurukshetra, India

- ◇ Awarded Visvesvaraya PhD Scheme Fellowship by the Ministry of Electronics & Information Technology, Govt. of India.

MASTER OF TECHNOLOGY

July 2011 - May 2013

Dayalbagh Educational Institute, Agra, India

- ◇ Awarded Graduate Aptitude Test Engineering (GATE) Fellowship by the Ministry of Human Resource & Development, Govt. of India.

MASTER OF SCIENCE

July 2009 - Dec 2010

Dayalbagh Educational Institute, Agra, India

- ◇ Recipient of the *Director Medal* for outstanding performance (First Rank) in the MSc Examination 2011.

GRANTS

1. Forecasting and Management of Cloud Workloads using Quantum Evolutionary Algorithm

Funding Agency: National Institute of Technology Tiruchirappalli, India

Completed

Role: Principal Investigator

Budget: ₹ 500000

2. Explaining Gender Gaps in Technology Creation: Evidence from the Indian Innovation System

Funding Agency: Indian Institute of Technology Indore, India

Sanctioned

Role: Co-Principal Investigator

Budget: ₹ 400000

PUBLICATIONS

Books

3. Jitendra Kumar, Ashutosh Kumar Singh, Anand Mohan, Rajkumar Buyya, "Machine Learning for Cloud Management", *Chapman & Hall*, 1st edition, 2021. (ISBN: 9780367626488) [[Link to Book](#)]
4. Abhishek Verma, Jitendra Kumar, Hari Mohan Gaur, Vrijendra Singh, Valentina E. Balas (eds.), "Advances in Cyber Security and Intelligent Analytics", *CRC Press*, 1st edition, 2022. (ISBN: 9781000821390) [[Link to Book](#)]
5. Jitendra Kumar, G. R. Gangadharan, Ashutosh Kumar Singh, Chung-Nan Lee (eds.), "Cloud of Things: Foundations, Applications, and Challenges", *Chapman & Hall*, 1st edition, 2024. (ISBN: 9781032484419) [[Link to Book](#)]
6. Vivek Kumar Singh, Jitendra Kumar, Deepika Saxena, Abhishek Verma, Thangarajah Akilan, Ashutosh Kumar Singh (eds.), "Machine Vision Analysis in Industry 5.0: Fundamentals, Applications, and Challenges", *Chapman & Hall*, 1st edition, 2025. (ISBN: 9781032874494) [[Link to Book](#)]

Book Chapters

7. Rohit Roushan Raj, Jitendra Kumar, Devidas S., "Heterogeneous Multi-Island Evolutionary Optimization Framework for Cloud Resource Utilization Forecasting", in *Metaheuristic Optimization for Social Good*, Elsevier, 2026. [**Accepted**]
8. Reshma MO, Jitendra Kumar, Abhishek Verma, "An Adaptive Deep Learning Approach for Stock Price Forecasting", in *Advances in Cyber Security and Intelligent Analytics*, CRC Press, 2022, pp. 1-12. [[Link to Publication](#)]
9. Harsh Rai, Vamshi Adouth, Deepika Saxena, Jitendra Kumar, "Cloud-Based Decentralized Freelancing Marketplace", in *Cloud of Things: Foundation, Applications, and Challenges*, Chapman & Hall, 2024, pp. 1-16. [[Link to Publication](#)]
10. Vinaytosh Mishra, Chandra Shekhar, Jitendra Kumar, Ashutosh Kumar Singh, "AI-Based Mass Screening of Tuberculosis Using Cloud-Based EHR and CXR - An Implication for India's National Strategic Plan to End TB", in *Cloud of Things: Foundation, Applications, and Challenges*, Chapman and Hall/CRC, 2024, pp. 1-10. [[Link to Publication](#)]

Journal Papers

11. Raseena T.P., Balasundaram S.R., Jitendra Kumar, "Graph-Based Feature Fusion Network with Multiscale Edge-Preserving Techniques for Polyp Identification", *Computers and Electrical Engineering*, vol. 130, Article 110867, 2026. (SCI/WoS, Q1, IF = 4.9) [[Link to Publication](#)]
12. Raseena T.P., Balasundaram S.R., Jitendra Kumar, "A Cognitive Dual-Attention Network with Feature Specificity for Automated CRC Polyp Detection", *Information Sciences*, vol. 725, Article 122727, 2026. (SCI/WoS, Q1, IF = 6.8) [[Link to Publication](#)]
13. Jitendra Kumar, Deepika Saxena, Kishu Gupta, Satyam Kumar, Ashutosh Kumar Singh, "A Comprehensively Adaptive Architectural Optimization-Ingrained Quantum Neural Network Model for Cloud Workloads Prediction", *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36 (10), pp. 19039-19053, 2025. (SCI/WoS, Q1, IF = 8.9) [[Link to Publication](#)]

14. Sushanshu Saurav, Jitendra Kumar, "Exploring the Effectiveness of Machine Learning Models on Livestock Behaviour Classification using 9-axial Inertial Sensor", *Evolutionary Intelligence*, vol. 18 (90), 2025. (Scopus/WoS, Q2, IF = 2.6) [[Link to Publication](#)]
15. Anjali, Vrijendra Singh, Jitendra Kumar, "GSSX: Golden Section-Based Symbolic Representation for Reducing Time Series Dimensionality", *International Journal of Data Science and Analytics*, vol. 20, pp. 5537-5554, 2025. (Scopus/WoS, Q2, IF = 2.8) [[Link to Publication](#)]
16. Jitendra Kumar, Deepika Saxena, Jatinder Kumar, Abhilash Bhadoria, Anjali, Ashutosh Kumar Singh, "A Hybrid Neural Network and Cooperative PSO Model for Dynamic Cloud Workloads Prediction", *Computing*, vol. 107, Article 72, 2025. (SCI/WoS, Q1, IF = 2.8) [[Link to Publication](#)]
17. Sakshi Chhabara, Jitendra Kumar, Deepika Saxena, Ashutosh Kumar Singh, "Classified Dynamic Hierarchical Load Balancer for Cloud Data Centers", *SN Computer Science*, vol. 6, Article 218, 2025. (Scopus) [[Link to Publication](#)]
18. Deepika Saxena, Niharika Singh, Kishu Gupta, Abhishek Verma, Vinaytosh Mishra, Jitendra Kumar, Ishu Gupta, Sakshi Chhabara, Rishabh Gupta, Jatinder Kumar, Ashutosh Kumar Singh, "An Intelligent Multi-Depot Vehicle Routing and Management Model for Smart Cities", *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 6, pp. 7740-7754, 2025. (SCI/WoS, Q1, IF = 8.4) [[Link to Publication](#)]
19. Jatinder Kumar, Deepika Saxena, Jitendra Kumar, Ashutosh Kumar Singh, Athanasios V. Vasilakos, "An Adaptive Evolutionary Neural Network model for Load Management in Smart Grid Environment", *IEEE Transactions on Network and Service Management*, vol. 22, pp. 242-254, 2024. (SCI/WoS, Q1, IF = 4.2) [[Link to Publication](#)]
20. Kishu Gupta, Deepika Saxena, Pooja Rani, Jitendra Kumar, Aaisha Makkar, Ashutosh Kumar Singh, "An Intelligent Quantum Cyber-Security Framework for Healthcare Data Management", *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 6884-6895, 2024. (SCI/WoS, Q1, IF = 5.9) [[Link to Publication](#)]
21. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "DeepCPD: Deep Learning with Vision Transformer for Colorectal Polyp Detection", *Multimedia Tools & Applications*, vol. 83, pp. 78183-78206, 2024. (SCI/WoS, Q1, IF = 3.00) [[Link to Publication](#)]
22. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "Exploring the Effectiveness of Deep Learning Architectures for Colorectal Polyp Detection: Performance Analysis and Insights", *SN Computer Science*, vol. 5, pp. 1-27, 2024. (Scopus) [[Link to Publication](#)]
23. Anjali, Hong-Nig Dai, Jitendra Kumar, Om Jee Pandey, "AERed: An Autoencoder-decoder Dimensionality Reduction Method for Wearable-Based Human Activity Recognition", *IEEE Sensors Journal*, vol. 23, pp. 29804-29814, 2023. (SCI/WoS, Q1, IF = 4.3) [[Link to Publication](#)]
24. Deepika Saxena, Jitendra Kumar, Ashutosh Kumar Singh, Stefan Schmid, "Performance Analysis of Machine Learning Centered Workload Prediction Models for Cloud", *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, pp. 1313-1330, 2023. (SCI/WoS, Q1, IF = 5.6) [[Link to Publication](#)]
25. Niharika Singh, Jitendra Kumar, Ashutosh Kumar Singh, Anand Mohan, "Privacy-preserving multi-keyword hybrid search over encrypted data in cloud", *Journal of Ambient Intelligence and Humanized Computing*, vol. 15, pp. 261-274, 2024. (SCI/WoS, Q1, IF = 3.662) [[Link to Publication](#)]
26. Jitendra Kumar, Ashutosh Kumar Singh, "Performance evaluation of metaheuristic algorithms for workload prediction in cloud environment", *Applied Soft Computing*, vol. 113, Part A, Article 107895, 2021. (SCI/WoS, Q1, IF = 7.2) [[Link to Publication](#)]

27. Deepika Saxena, Ishu Gupta, Jitendra Kumar, Ashutosh Kumar Singh, Xiaoqing Wen, "A Secure and Multi-objective Virtual Machine Placement Framework for Cloud DataCenter", *IEEE Systems Journal*, vol. 16, no. 2, pp. 3163-3174, 2022. (SCI/WoS, **Q1**, IF = 4.0) [[Link to Publication](#)]
28. Ashutosh Kumar Singh, Deepika Saxena, Jitendra Kumar, Vrinda Gupta, "A quantum approach towards the adaptive prediction of cloud workloads", *IEEE Transactions on Parallel and Distributed Systems*, vol. 32, no. 12, pp. 2893-2905, 2021. (SCI/WoS, **Q1**, IF = 5.6) [[Link to Publication](#)]
29. Jitendra Kumar, Ashutosh Kumar Singh, Rajkumar Buyya, "Self directed learning based workload forecasting model for cloud resource management", *Information Sciences*, vol. 543, pp. 345-366, 2021. (SCI/WoS, **Q1**, IF = 8.1) [[Link to Publication](#)]
30. Jitendra Kumar, Ashutosh Kumar Singh, Rajkumar Buyya, "Ensemble learning based predictive framework for virtual machine resource request prediction", *Neurocomputing*, vol. 397, pp. 20-30, 2020. (SCI/WoS, **Q1**, IF = 5.5) [[Link to Publication](#)]
31. Jitendra Kumar, Ashutosh Kumar Singh, "Decomposition based cloud resource demand prediction using extreme learning machines", *Journal of Network and Systems Management*, vol. 28, pp. 1775-1793, 2020. (SCI/WoS, **Q2**, IF = 4.1) [[Link to Publication](#)]
32. Jitendra Kumar, Deepika Saxena, Ashutosh Kumar Singh, Anand Mohan, "BiPhase adaptive learning-based neural network model for cloud datacenter workload forecasting", *Soft Computing*, vol. 24, pp. 14593-14610, 2020. (SCI/WoS, **Q2**, IF = 3.1) [[Link to Publication](#)]
33. Jitendra Kumar, Ashutosh Kumar Singh, Anand Mohan, "Resource-efficient load-balancing framework for cloud data center networks", *ETRI Journal*, vol. 43, pp. 53-63, 2021. (SCI/WoS, **Q2**, IF = 1.3) [[Link to Publication](#)]
34. Jitendra Kumar, Ashutosh Kumar Singh, "Cloud datacenter workload estimation using error preventive time series forecasting models", *Cluster Computing*, vol. 23, iss. 2, pp. 1363-1379, 2020. (SCI/WoS, **Q1**, IF = 3.6) [[Link to Publication](#)]
35. Jitendra Kumar, Ashutosh Kumar Singh, "Performance assessment of time series forecasting models for cloud datacenters' workload prediction", *Wireless Personal Communications*, vol. 116, pp. 1949-1969, 2021. (SCI/WoS, **Q2**, IF = 1.9) [[Link to Publication](#)]
36. Jitendra Kumar, Ashutosh Kumar Singh, "Adaptive learning based prediction framework for cloud datacenter networks' workload anticipation", *Journal of Information Science and Engineering*, vol. 36, iss. 5, pp. 981-992, 2020. (SCI/WoS, **Q2**, IF = 1.142) [[Link to Publication](#)]
37. Ashutosh Kumar Singh, Jitendra Kumar, "Secure and energy aware load balancing framework for cloud data centre networks", *Electronics Letters*, vol. 55, iss. 9, pp. 540-541, May, 2019. (SCI/WoS, **Q2**, IF = 0.7) [[Link to Publication](#)]
38. Jitendra Kumar, Ashutosh Kumar Singh, "An efficient machine learning approach for virtual machine resource demand prediction", *International Journal of Advanced Science and Technology*, vol. 123, pp. 21-30, February, 2019. (Scopus) [[Link to Publication](#)]
39. Jitendra Kumar, Ashutosh Kumar Singh, "Workload prediction in cloud using artificial neural network and adaptive differential evolution", *Future Generation Computer Systems*, vol. 81, pp. 41-52, April, 2018. (SCI/WoS, **Q1**, IF = 6.2) [[Link to Publication](#)]
40. Jitendra Kumar, Rimsha Goomer, Ashutosh Kumar Singh, "Long Short Term Memory Recurrent Neural Network (LSTM-RNN) based workload forecasting model for cloud datacenters", *Procedia Computer Science*, vol. 125, pp. 676-682, January, 2018. (Scopus) [[Link to Publication](#)]

41. Rimsha Alam, Usha Chouhan, Jitendra Kumar, "From Multi-Omics Integration to Digital Twin: Machine and Deep Learning-Driven Frameworks for Precision Oncology", *IEEE Reviews in Biomedical Engineering*. (SCI/WoS, **Q1**) In Review since 25/03/2026
42. Kishu Gupta, Deepika Saxena, Jitendra Kumar, Ashutosh Kumar Singh, Chung-Nan Lee, "A Quantum Intelligence Model for Proactive Data Breach Mitigation in Healthcare Environments", *IEEE Transactions on Automation Science and Engineering*. (SCI/WoS, **Q1**) Revision Submitted on 20/11/2025
43. Niharika Singh, Kishu Gupta, Rishabh Gupta, Deepika Saxena, Abhishek Verma, Jitendra Kumar, Ashutosh Kumar Singh, "Learning-based Vehicle Routing: A Survey for AI/ML-driven Sustainable Urban Mobility", *IEEE Intelligent Transportation Systems Transactions*. (SCI/WoS, **Q1**) In Review since 24/10/2025
44. Deepika Saxena, Kishu Gupta, Jitendra Kumar, Jatinder Kumar, Sakshi Chhabara, Vinaytosh Mishra, Ashutosh Kumar Singh, "Indian Wedding System Optimization (IWSO): A Novel Socially Inspired Metaheuristic with Operational Design and Analysis", *Applied Soft Computing*. (SCI/WoS, **Q1**) In Review since 28/05/2026, [[Link to Preprint](#)]
45. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "SCAR-Net: Sobel-filtered Convolutional Block Attention Module-Driven Residual Network for Improved Colon Polyp Classification", *Engineering Applications of Artificial Intelligence*. (SCI/WoS, **Q1**) In Review since 26/02/2025
46. Rishabh Gupta, Deepika Saxena, Sakshi Chhabara, Jitendra Kumar, Ashutosh Kumar Singh, Stefan Schmid, "Maclaurin Series based Deep Neural Network Model for Privacy-Preserving Data in Cloud Infrastructure", *IEEE Transactions on Services Computing*. (SCI/WoS, **Q1**) In Review since 31/03/2024
47. Billy Pik Lik Lau, Terence Peng Lian Tan, Ashutosh Kumar Singh, Jitendra Kumar, "A Knapsack based Dependence Relationship Validation Mechanism for Verifying Agents' Social Status", *Computing*. (SCI/WoS, **Q1**) Revision Submitted on 22/07/2025

Conference Papers

48. Rimsha Alam, Usha Chouhan, Jitendra Kumar, "A Forensic Multi-Omics Framework for Robust Breast Cancer Biomarker Discovery Using Explainable AI and Network Biology", *IEEE International Conference on Artificial Intelligence in Engineering and Technology 2026, Malaysia, 2026*. (Scopus) [Submitted on 31/07/2026]
49. Rohit Roushan Raj, Jitendra Kumar, Devidas S., "Collaborative Learning Enabled Workload Prediction Model for Large-Scale Cloud Environments", *TENCON 2026, Indonesia, 2026*. (Scopus) [**Accepted** for presentation]
50. Amit Singh Kushwaha, Jitendra Kumar, Madhvi Shakya, "A Privacy Aware Data Driven Approach to Predict Necessary Bolus Events in Type-1 Diabetes using LSTM Architecture", *First International Conference on Statistics, Optimization, and Machine Learning, MANIT Bhopal, India, 2026*. (Scopus) [**Accepted & Presented**]
51. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "A Visionary Approach for Enhanced CRC Diagnosis by Leveraging Visual Geometry Group's Classic Deep Network", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, Bhopal, India, 2024*, pp. 964-971. (Scopus) [[Link to Publication](#)]
52. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "A Visionary Approach for Enhanced CRC Diagnosis by Leveraging Visual Geometry Group's Classic Deep Network", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, IIIT Bhopal, India, 2024*, pp. 964-971. (Scopus) [[Link to Publication](#)]

53. Tushar Prasanna Swaminathan, Christopher Silver, Jitendra Kumar, Thangarajah Akilan, "Benchmarking Deep Learning Models on NVIDIA Jetson Nano for Real-Time Systems: An Empirical Investigation", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, IIIT Bhopal, India, 2024*, pp. 906-913. (Scopus) [[Link to Publication](#)]
54. P. Ravi Kumar, Aman Wakade, Jatinder Kumar, Smruti Rekha Swain, Jitendra Kumar, Ashutosh Kumar Singh, "On Detecting Frauds in Retail Transactions for Enhanced Security with Deep Learning Model", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, IIIT Bhopal, India, 2024*, pp. 468-475. (Scopus) [[Link to Publication](#)]
55. Raseena T.P., Aditya Kumar Soni, Jitendra Kumar, Thangarajah Akilan, Ashutosh Kumar Singh, "An Analysis of YOLOvX Deep Learning Models for Colon Cancer Detection", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, IIIT Bhopal, India, 2024*, pp. 81-90. (Scopus) (Best Paper Presentation Award to Dr. Raseena T.P.) [[Link to Publication](#)]
56. Sanjeev Kumar, Yudhishtira Sapru, Jitendra Kumar, Sugandha Sapru, "Image analysis and fine-tuned ResNet50 model for effective malware detection", *7th International Conference on Recent Trends in Image Processing and Pattern Recognition, IIIT Bhopal, India, 2024*, pp. 1098-1106. (Scopus) [[Link to Publication](#)]
57. Keerthana Senthilnathan, Jitendra Kumar, "Cloud resource usage forecasting using a deep learning-based predictive framework", *3rd International Conference on Advanced Communication and Intelligent Systems (ICACIS), Jawaharlal Nehru University, New Delhi, India, 2024*. [Accepted & Presented]
58. Raseena T.P., Jitendra Kumar, Balasundaram S.R., "A Residual Learning Approach Towards the Diagnosis of Colorectal Disease Effectively", *2023 6th International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R), University of Derby, United Kingdom, 2023*, pp. 160-172. (Scopus) [[Link to Publication](#)]
59. Sajeev Vijaykumar, Jitendra Kumar, "Cloud Resource Usage Forecasting Using Neural Network and Artificial Lizard Search Optimization", *2022 10th IEEE Humanitarian Technology Conference (R10 HTC), Hyderabad, India, 2022*, pp. 7-12. (Scopus) [[Link to Publication](#)]
60. Jitendra Kumar, Ashutosh Kumar Singh, "Cloud resource demand prediction using differential evolution based learning", *2019 7th International Conference on Smart Computing & Communications (ICSCC), Sarawak, Malaysia, 2019*, pp. 1-5. (Scopus) (Best Paper Award) [[Link to Publication](#)]
61. Vartika Sharma, Sizman Kaur, Jitendra Kumar, Ashutosh Kumar Singh, "A fast parkinson's disease prediction technique using PCA and artificial neural network", *2019 International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2019*, pp. 1491-1496. (Scopus) [[Link to Publication](#)]
62. Sukhman Singh, Tarun Kumar Madan, Jitendra Kumar, Ashutosh Kumar Singh, "Stock market forecasting using machine learning: today and tomorrow", *Second International Conference on Intelligence Computing, Instrumentation and Control Technologies (ICICT-2019), Kannur, Kerala, India, 2019*, pp. 738-745. (Scopus) [[Link to Publication](#)]
63. Preetesh K. Yadav, Sourav Pareek, Shaif Shakeel, Jitendra Kumar, Ashutosh Kumar Singh, "Advancements and security issues of IoT & cyber physical systems", *2019 International Conference on Intelligent Computing and Control Systems (ICICCS), Madurai, India, 2019*, pp. 940-945. (Scopus) [[Link to Publication](#)]
64. Jitendra Kumar, Ashutosh Kumar Singh, "Dynamic resource scaling in cloud using neural network and black hole algorithm", *2016 Fifth International Conference on Eco-friendly Computing and Communication Systems (ICECCS), Bhopal, India, 2016*, pp. 63-67. (Scopus) (Best Paper Award) [[Link to Publication](#)]

65. Mayank Dixit, Jitendra Kumar, Rajesh Kumar, "Internet of things and its challenges", 2015 *International Conference on Green Computing and Internet of Things (ICGCIoT)*, Noida, India, 2015, pp. 810-814. (Scopus) [[Link to Publication](#)]
66. Jitendra Kumar, Rajesh Kumar, Mayank Dixit, "Result merging in meta-search engine using genetic algorithm", *International Conference on Computing, Communication & Automation (ICCCA)*, Noida, India, 2015, pp. 299-303. (Scopus) [[Link to Publication](#)]
67. Jitendra Kumar, Lotika Singh, Sandeep Paul, "GPU based parallel cooperative Particle Swarm Optimization using C-CUDA: A case study", 2013 *IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, Hyderabad, India, 2013, pp. 1-8. (Scopus) [[Link to Publication](#)]

Technical Reports

68. Sakshi Chhabara, Deepika Saxena, Jitendra Kumar, Ashutosh Kumar Singh, "Classified Dynamic Hierarchical Load Balancer for Cloud Data Centers", *PREPRINT (Version 1) available at Research Square*, 2022. [[Link to Publication](#)]
69. Pooja Tiwari, Simran Mehta, Nishtha Sakhuja, Jitendra Kumar, Ashutosh Kumar Singh, "Credit Card Fraud Detection using Machine Learning: A Study", *arXiv:2108.10005*, 2021. [[Link to Publication](#)]
70. Harsh Mittal, Deepak Rikhari, Jitendra Kumar, Ashutosh Kumar Singh, "A study on Machine Learning Approaches for Player Performance and Match Results Prediction", *arXiv:2108.10125*, 2021. [[Link to Publication](#)]

STUDENT SUPERVISION¹

	Ph.D.	Masters[†]	Undergraduates	Internships	Total
Ongoing	5	3/3/0	2	0	13
Completed	1	5/44/7	0	5	62

[†] Includes M.Tech./M.Sc./MCA

AWARDS & RECOGNITIONS

1. Best Paper Presentation Award [2025]
2025 International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R), IIIT Bhopal, India (Co-authored Paper)
2. Best PhD Thesis Award [2023]
2023 International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R), University of Derby, United Kingdom
3. Best Paper Award [2019]
2019 International Conference on Smart Computing & Communication (ICSCC), Curtin University, Malaysia
4. Best Paper Award [2016]
2016 International Conference on Eco-friendly Computing & Communication Systems (ICECCS), MANIT Bhopal, India
5. Visvesvaraya PhD Scheme Fellowship [October 2015 - July 2019]
Ministry of Electronics & Information Technology, Government of India
6. GATE Fellowship [July 2011 - May 2013]
Ministry of Human Resource Development, Government of India
7. Director's Medal [2011]
M.Sc. Examination 2011, Dayalbagh Educational Institute, Agra, India

¹Complete list is available at Page 12.

EDITORIALS

1. Diana Berbecaru, Riccardo Sisto, Deepika Saxena, Jitendra Kumar, Cybersec Attacks and Defenses, Computer Networks, Elsevier, 2025 [\[Link\]](#)
2. Ashutosh Kumar Singh, Jitendra Kumar, Deepika Saxena, K C Santosh, Revolutionizing Healthcare Informatics with Generative Artificial Intelligence: Innovations and Implications, IEEE Journal of Biomedical and Health Informatics, 2024 [\[Link\]](#)
3. Ashutosh Kumar Singh, K C Santosh, Jitendra Kumar, Deepika Saxena, Aaisha Makkar, Proceedings of 7th International Conference on Recent Trends in Image Processing and Pattern Recognition, Procedia Computer Science, 2024 [\[Link\]](#)
4. Vivek Kumar Singh, Jitendra Kumar, Deepika Saxena, Abhishek Verma, Thangarajah Akilan, Ashutosh Kumar Singh, Machine Vision Analysis in Industry 5.0: Fundamentals, Applications, and Challenges, CRC Press, 2025 [\[Link\]](#)
5. Ashutosh Kumar Singh, Jitendra Kumar, Deepika Saxena, Athanasios V. Vasilakos, eXplainable Artificial Intelligence (XAI): Methods, Applications, and Challenges, Computers and Electrical Engineering, Elsevier, 2024 [\[Link\]](#)
6. Jitendra Kumar, G. R. Gangadharan, Ashutosh Kumar Singh, Chung-Nan Lee, Cloud of Things: Foundations, Applications, and Challenges, CRC Press, 2024 [\[Link\]](#)
7. Jonathan Wu, Thangarajah Akilan, Jitendra Kumar, Chengsheng Yuan, Machine Vision Applications and Efficient Deep Learning Models for Resource-Limited Learning, Journal of Imaging, MDPI, 2022 [\[Link\]](#)
8. Abhishek Verma, Jitendra Kumar, Hari Mohan Gaur, Vrijendra Singh, Valentina E. Balas, Advances in Cyber Security and Intelligent Analytics, CRC Press, 2022 [\[Link\]](#)

TALKS & PRESENTATIONS

1. Emerging Research Opportunities and Practical Applications in Machine Learning
Dayalbagh Educational Institute, Agra, India [March 19, 2026]
2. Predictive Data Analysis and Intelligent System Design
Dayalbagh Educational Institute, Agra, India [August 11, 2025]
3. Neural Networks and Forecasting
Indian Institute of Information Technology & Management, Gwalior, India [May 6, 2025]
4. Education and Artificial Intelligence
Mahatma Hansraj Malaviya Mission Teacher Training Centre (MH-MMTTC), Hansraj College, University of Delhi [December 28, 2024]
5. Revolutionizing Education through Artificial Intelligence
Hansraj College, University of Delhi [July 7, 2023]
6. Machine Learning in the Air: Today and Tomorrow
VIT Vellore [May 9, 2023]
7. Neural Networks for Predictive Modelling
VIT Chennai [March 16, 2023]

8. Presenting Scientific Data: Plots, Postscripts, and PDF Graphics in $\text{\LaTeX}$
NIT Kurukshetra [March 15, 2022]
9. Quantum Evolutionary Neural Network: Bio-inspired computing and quantum principles
ABESIT College of Engineering, Ghaziabad, Uttar Pradesh, India [November 25, 2021]
10. Research Writing and Document Preparation using $\text{\LaTeX}$
NIT Kurukshetra [August 2021]
11. Evolutionary Neural Networks
PSNA College of Engineering & Technology, Dindigul, Tamil Nadu, India [January 14, 2021]
12. Plots & Figures in $\text{\LaTeX}$
ABESIT College of Engineering, Ghaziabad, Uttar Pradesh, India [February 11, 2021]
13. Machine Learning: Today and Tomorrow
Gulzar Group of Institutes, Ludhiana, Punjab, India [October 6, 2020]
14. A Residual Learning Approach Towards the Diagnosis of Colorectal Disease Effectively
2023 International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R)
University of Derby, United Kingdom [December 7-8, 2023]
15. Advancements and Security Issues of IoT & Cyber Physical Systems
2017 International Conference on Smart Computing & Communications (ICSCC)
National Institute of Technology, Kurukshetra, India [December 7-8, 2017]
16. Dynamic Resource Scaling in Cloud using Neural Network and Black Hole Algorithm
2016 Fifth International Conference on Eco-friendly Computing & Communication Systems (ICECCS)
National Institute of Technology, Bhopal, India [December 8-9, 2016]
17. Result Merging in Meta-search Engine using Genetic Algorithm
2015 International Conference on Computing, Communications & Automation (ICCCA)
Galgotias University, India [May 15-16, 2015]
18. GPU based Parallel Cooperative Particle Swarm Optimization using C-CUDA: A Case Study
2013 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)
Hyderabad International Convention Center (HICC), Hyderabad, India [July 7-10, 2013]

PROFESSIONAL MEMBERSHIPS

Senior Member

Institute of Electrical and Electronics Engineers (IEEE) [August 2012 - Present]

Member

IEEE Computational Intelligence Society [December 2021 - Present]

Association for Computing Machinery (ACM) [January 2021 - Present]

Machine Intelligence Research Labs (MIR Labs) [July 2020 - Present]

IEEE Young Professionals [January 2014 - Present]

IEEE Cloud Computing Community [March 2013 - Present]

IEEE Communications Society [January 2013 - December 2013]

IEEE SIGHT [January 2021 - Present]

CONFERENCES/PROGRAMS ORGANIZED

1. *Second International Conference on Advances in Artificial Intelligence* [Program Chair]
IIIT Bhopal, India & Vizja University, Poland 2026
2. *FDP on Scientific Writing and $\text{\LaTeX}$ for Research Publications* [Coordinator]
MANIT Bhopal, India 24/11/2025 - 28/11/2025
3. *First International Conference on Advances in Artificial Intelligence* [Program Chair]
Gulf Medical University, UAE & IIIT Bhopal, India 13/11/2025 - 14/11/2025
4. *STTP on $\text{\LaTeX}$ Programming for Technical Writing & Presentation* [Coordinator]
MANIT Bhopal, India 07/07/2025 - 11/07/2025
5. *7th Intl. Conf. on Recent Trends in Image Processing and Pattern Recognition* [Program Chair]
IIIT Bhopal, India & MANIT Bhopal, India 19/12/2024 - 20/12/2024
6. *Special Session on "Advanced Data Science and Machine Learning in Cutting-Edge Applications with Security Measures," 3rd International Conference on Advanced Communication and Intelligent Systems* [Session Chair]
Jawaharlal Nehru University, India 2024
Session Chairs: Anil Kumar Sagar, Jitendra Kumar
7. *Special Session on "Cyber Medical System: Personalized Healthcare," 6th International Conference on Recent Trends in Image Processing and Pattern Recognition* [Session Chair]
University of Derby, United Kingdom 07/12/2023 - 08/12/2023
Session Chairs: Jitendra Kumar, Ashutosh Kumar Singh, Chung-Nan Lee

VOLUNTEER ACTIVITIES

1. Sixth International Conference on Smart Computing and Communication (ICSCC-2017)
2. National IT Challenge for Youth with Disabilities 2016

COMPETITIONS PARTICIPATED

1. DExtreme, 6.0 IEEE Xtreme Programming Competition 2012
2. Quantos, 11.0 IEEE Xtreme Programming Competition 2017

COMPLETE LIST OF SUPERVISED STUDENTS

PhD Students

					?
					?
SN	Student Name	Registration Year	Research Area/ Thesis Title	Status	
					?
1	Sanjana Prajapati	2025-2026	Smart Grid Management	Ongoing	
2	Ruchi Shrivastava	2025-2026	Machine Learning	Ongoing	
3	Sanchita Shrivastava	2025-2026	Cyber Security	Ongoing	
4	Rohit Roushan Raj	2024-2025	Federated Learning	Ongoing	
5	Rimsha Alam	2024-2025	Deep Learning Approaches for Multi-Omics based Disease Detection	Ongoing	
6	Dr. Raseena T.P.	2020-2021	Improving Performance in Colorectal Polyp Classification using Efficient Deep Learning Approaches	Awarded	
					?
					?

Masters Students

					?
					?
SN	Student Name	Dissertation Title	Course/Year		
					?
7	Stuti D Tiwari	Deepfake Detection	M.Tech(ASE)/2027		
8	Gokul Raj NV	---	M.Tech(ASE)/2027		
9	Mridul Kumar Thepey	---	M.Tech(ASE)/2027		
10	Kunal Burande	Cloud Workload Forecasting in Large Scale Clusters : A Federated Learning Approach with Bio Inspired Aggregation	MCA/2027		
11	Kartik Yadav	Cyber Attack Detection using Marine Predator Algorithm	MCA/2027		
12	Soumallya Naskar	Accident Severity Prediction Using Ensemble Machine Learning with Harris Hawks Optimization	MCA/2027		
13	Amit Kumar Kushwaha	Collaborative Learning Enabled Personalized Monitoring for Type 1 Diabetes	M.Tech(ASE)/2026		
14	Yash Vishwakarma ^{*,L}	Legacy Migration, Cloud-Native Scalability, And Enterprise Ai Systems	M.Tech(ASE)/2026		
15	Tardej Tudul ^L	Customer Churn Prediction	MCA/2026		
16	Mansi Birla ^L	Traffic Migration	MCA/2026		
17	Preeti Gautam ^L	Multi-Model Autonomous Target Identification System	MCA/2026		
18	Lokendra Mahale ^L	News Publishing Platform	MCA/2026		
19	Manas Das ^L	Data Augmentation & Latent Space Interpolation	MCA/2026		
20	Senha C. Dange ^L	Kerna Client Operations & Workflow Governance Platform	MCA/2026		
21	Yashwant Bharti ^L	Enhancing Long-Term Cloud Workload Forecasting Framework	MCA/2026		
22	Md Reyaj ^L	Smart Manufacturing Monitoring And Defect Detection System	MCA/2026		

23	Manish Verma ^ℓ	Design And Development Of A Modern E-Commerce: Brilliant Photo Group Singapore	MCA/2026
24	Avlok Raj ^ℓ	Cards System: Data Centre Operations	MCA/2026
25	Rohit Kumar ^ℓ	Micro Peer-To-Peer Lending Platform	MCA/2026
26	Praveen Kumar Arya ^ℓ	Sps Erp System	MCA/2026
27	Vinay Sharma ^ℓ	Job Hr Dashboard	MCA/2026
28	Saswat Moharana ^ℓ	Job Hive- Job Portal Application	MCA/2026
29	Vicky Verma ^ℓ	Loan Management System	MCA/2026
30	Risu Gupta	Time-Series Based Cpu And Memory Usage Prediction Using Collaborative Evolutionary Optimization	MCA/2026
31	Priya Tripathi ^ℓ	Hybrid Personalized Recommendation System With Llm Integration In E-Commerce	MCA/2026
32	Mohit Aman ^ℓ	Performance Testing And Quality Analysis	MCA/2025
33	Priyanka Prakash Sahu ^ℓ	Interior Design Website	MCA/2025
34	Shivam Gupta ^ℓ	Secret Management	MCA/2025
35	Kirti Prakash Bhavsar [†]	Cloud Workload Forecasting for Improved Resource Utilization: A Differential Evolution-Neural Network Hybrid Approach	MSc/2024
36	Shrestha Bhattacharjee [†]	NeuralRoute: Graph-Based Traffic Route Optimization for Efficient Urban Mobility	MSc/2024
37	Ashutosh Wardhan [†]	An Evolutionary Neural Network Approach for Cloud Workload Forecasting	MCA/2024
38	Bikash Kumar Yadav [†]	Colorectal Cancer Detection Using ML and DL Algorithms	MCA/2024
39	Deepak Choudhary [†]	Artificial Neural Network for Detecting Credit Card Fraud	MCA/2024
40	Kratika Patidar [†]	Machine Learning Algorithms for Credit Card Fraud Detection	MCA/2024
41	Lishu Gupta [†]	Automated Attendance Management Using Wireless Technologies	MCA/2024
42	Raja Kumar Thakur [†]	Fuzzy Time Series Forecasting	MCA/2024
43	Shivam Engla [†]	Medical Image Denoising Using Convolutional Autoencoder	MCA/2024
44	Sumit Kushwaha [†]	Balancing and Generating medical images using Deep Learning	MCA/2024
45	Utpal Tripathi [†]	Software Aging Prediction Using Deep Learning Methods	MCA/2024
46	Satyam Kumar	Automatic Quantum Neural Network Architecture Learning for Cloud Workload Forecasting	MTech/2023
47	Akhil Kumar	Accelerating FCNN training through Memory Off-loading with Accelerator	MTech/2023
48	Rakesh Kumar	A Novel Meta-heuristic Optimization Algorithm for Cloud Resource Prediction	MSc/2023
49	Sudhanshu Saurav	Predictive Analysis of Cattle Monitoring using 9-Axial Inertial Sensor Data	MSc/2023
50	Abhilash Bhadoriya	Cooperative Neural Network for Workload Prediction in Cloud Datacenter	MCA/2023
51	Aditya Kumar Soni	Detecting Colon Cancer: A Deep Learning Approach using YOLOvX	MCA/2023
52	Harsh Rai	Optimal Workload Placement in Cloud-Fog Environment: A Gateway-Driven Approach	MCA/2023
53	Harsh Sharma	Image Classification using Deep Learning Methods	MCA/2023
54	Neetu Pandint	Cloud Resource Workload Forecasting using Swarm Intelligence	MCA/2023
55	Nikita Das ^ℓ	Config Service: A Service Based Web Application	MCA/2023

56	Sangam Raja	DepressionSense: Leveraging Ensemble LSTM for Depression Detection	MCA/2023
57	Sanskar Dhanotiya	Object Detection Using Deep Learning	MCA/2023
58	Shivangi Gupta ^ℓ	Immersive Maintenance: A Mixed Reality Application Collins Aerospace	MCA/2023
59	Nikhil Kumar	Automatic Colon Polyp Detection: An Advanced Deep Learning Approach	MTech/2022
60	Reshma MO	Stock Market Forecasting using Machine Learning	MSc/2022
61	Amit Kumar Agrawal	Tamil Voice to English Voice Translation	MSc/2022
62	Darshini Gupta	Text Detection from Images using Machine Learning	MCA/2022
63	Akhilesh Kumar	Sign Language Recognizer using Deep Learning	MCA/2021
64	Mahesh Kumar S ^ℓ	Compare the performance of Hand-Made ML and Auto-ML	MCA/2021
65	Amit Sahu ^{*,ℓ}	SAP S/4 HANA Extension Business Logic	MCA/2021
66	Hardik Jain ^{*,ℓ}	Nitty-Gritty of Campaign Implementation - Payment Trigger LOP - Fee Waiver	MCA/2021
67	Mayur Gehlod ^{*,ℓ}	Handwritten Text Recognition	MCA/2021
68	Mantu Kumar Gupta [*]	Natural Disasters Detection using Deep Learning and Computer Vision	MSc/2021

*Co-supervised Dissertation

^ℓIndustrial Dissertation

[†]Partially Supervised Dissertation

Undergraduate Students

SN	Student Name	Dissertation Title	Course/Year
69	Prachi Soni	Electricity Load Forecasting using Equilibrium Optimization	B.Tech./2027
70	Ch. Madhusudan	Federated Machine Learning Approach for Cloud Workload Forecasting	B.Tech./2027

Research Interns

SN	Student Name	Institute	Topic	Year
71	Gaurav Jaiswal	National Institute of Technology Tiruchirappalli, India	Federated Learning	2024-2025
72	Harsh Kumar Singh	National Institute of Technology Tiruchirappalli, India	Federated Learning	2024-2025
73	Keerthana S.	National Institute of Technology Tiruchirappalli, India	Deep Learning	2022-2023
74	Sanjeev Vijaykumar	National Institute of Technology Tiruchirappalli, India	Computational Intelligence	2021-2022

^α funded by German Federal Ministry of Education and Research under DAAD RISE 2021